

SmartAudit Pro - User Manual

Audit Sample Selector

1. System Requirements

Windows/macOS/Linux desktop with Python 3.10 or later. Internet access may be required on first run if Python packages are not already installed.

2. Installation

1. Download or clone the GitHub repository.
2. Open a terminal/command prompt in the project folder.
3. Run: `pip install -r requirements.txt`
4. Run: `python SmartAudit_Pro.py`

3. Demonstration Setup

Use `sample_data/SmartAudit_Demo_Population.xlsx`. Recommended demonstration parameters: threshold ₹10,00,000; sample size 25; seed 42.

4. Loading a Population

5. Open the Population & Sampling tab.
6. Click Browse.
7. Select an Excel or CSV file.
8. For Excel, the application reads the first worksheet.
9. Confirm the field mappings for Amount, Date, User and Narration.

5. Sampling Methods

Method	How to demonstrate
Random	Choose Random, sample size 25, seed 42, then click Select Samples.
Systematic	Choose Systematic and click Select Samples.
Stratified	Choose Stratified and click Select Samples.
Risk-based	Choose Risk-based and click Select Samples. Review Risk Score and Risk Reasons.

6. Dashboard

Open the Dashboard tab after loading a population. The left chart shows the absolute transaction-value distribution. The right chart shows Low, Medium and High categories based on the prototype's rule-based risk score.

7. Sample-size Calculator

Enter population size, confidence level, tolerable deviation and expected deviation, then click Calculate. Treat the output as an educational planning indicator only.

8. Exporting Reports

After selecting samples, click Export Excel Report or Export PDF Report. The Excel report contains summary information, selected samples, methodology notes and a risk chart. The PDF provides a concise summary and up to the first 100 selected items.

9. Recommended Capstone Demo Sequence

1. Load the 500-record demo workbook.
2. Explain population KPIs.
3. Run Random sampling.
4. Run Systematic sampling.
5. Run Stratified sampling.
6. Run Risk-based selection and explain the distinction from statistical sampling.
7. Show Dashboard.
8. Show Sample-size Calculator.
9. Export Excel and PDF reports.
10. Close with limitations and professional-judgement statement.

10. Troubleshooting

Issue	Action
Application does not start	Confirm Python 3.10+ and install requirements.txt.
Excel does not load	Use .xlsx/.xlsm and ensure the first worksheet contains headers in row 1.
Wrong amount mapping	Select the correct Amount field from the dropdown.
Date risk rules do not trigger	Use recognised date formats or Excel date

	values supported by the imported workbook.
Report button warns	A sample must be selected before exporting.
Package installation fails	Check internet access and permissions, then run <code>pip install -r requirements.txt</code> manually.

11. Audit-use Disclaimer

SmartAudit Pro is an academic prototype. Before any real audit use, the auditor should validate the population, sampling design, parameters, rule logic and outputs. The application does not replace compliance with applicable Standards on Auditing or professional judgement.